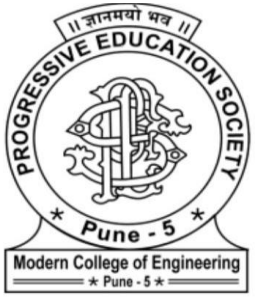
AI-Driven Predictive Maintenance System for Industrial Equipment

(An AIML Domain Project)

Submitted by,
Vedant Bhaskar Porje (PRN: 72312326J)
Under the Guidance of
Prof. Mrs. Mayuri A. Khade

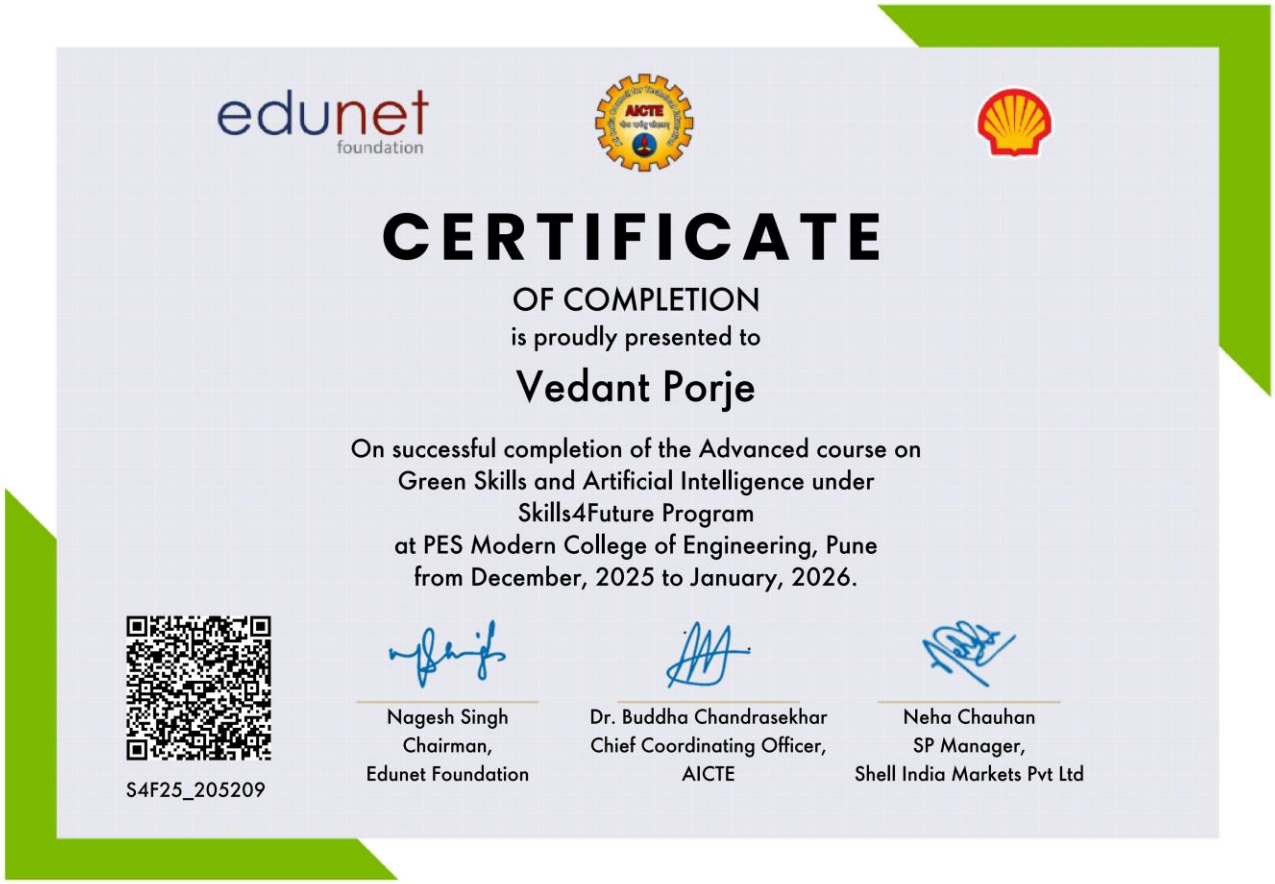


PES Modern College of Engineering
Department of Computer Engineering
Savitribai Phule Pune University
25-February-2026

Outline

1. Internship Completion Certificate
2. Introduction
3. Problem Statement
4. Objectives
5. Scope
6. Project Limitations
7. Dataset Description
8. Technological Details
9. Architecture Diagram
10. Feature Engineering Strategy
11. Model Performance & Evaluation
12. Anamoly Detection & Early Warning Analysis
13. Result
14. Conclusion
15. Future Scope\
16. Research and References

Internship Completion Certificate



edunet foundation

AICTE

CERTIFICATE

OF COMPLETION

is proudly presented to

Vedant Porje

On successful completion of the Advanced course on
Green Skills and Artificial Intelligence under
Skills4Future Program
at PES Modern College of Engineering, Pune
from December, 2025 to January, 2026.

S4F25_205209

Nagesh Singh
Chairman,
Edunet Foundation

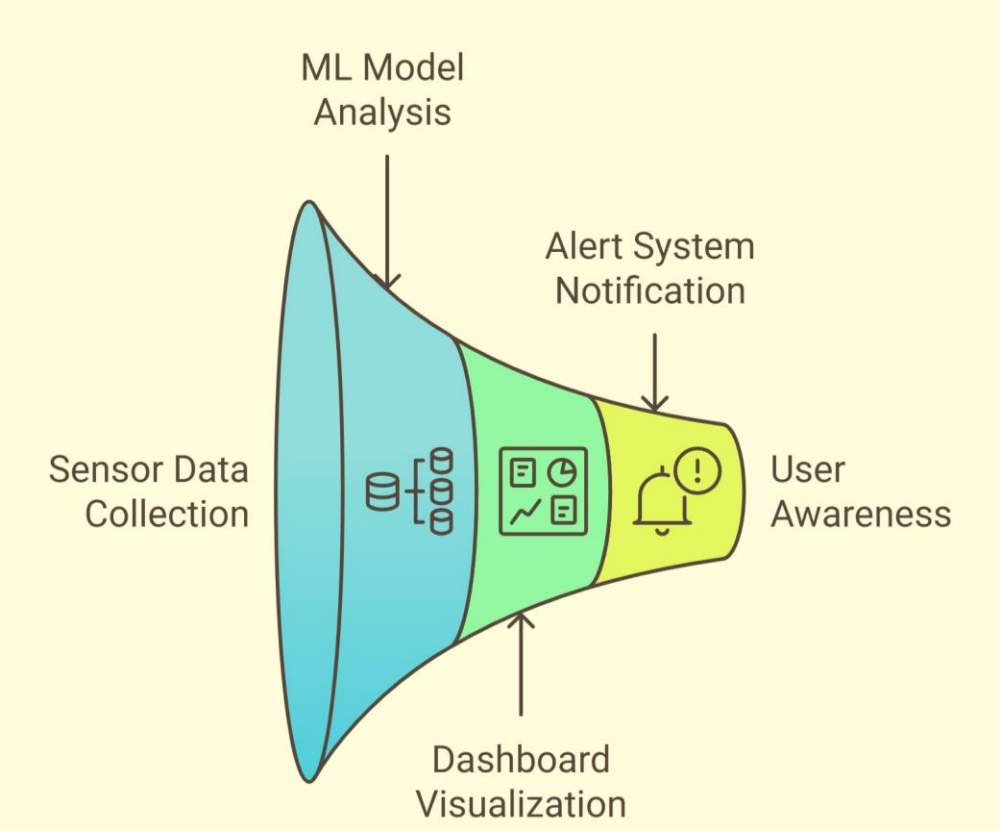
Dr. Buddha Chandrasekhar
Chief Coordinating Officer,
AICTE

Neha Chauhan
SP Manager,
Shell India Markets Pvt Ltd

Introduction

Unplanned equipment failures cause significant financial losses in industrial environments, and traditional maintenance strategies — whether reactive or scheduled — remain inefficient and costly. The growing availability of **industrial sensor data** offers a better path forward, enabling **real-time monitoring** of equipment health. By leveraging machine learning to identify degradation patterns, this project develops an end-to-end AI system that predicts **Remaining Useful Life (RUL)** and detects anomalies early, integrated with an interactive dashboard for real-time visibility and informed maintenance decisions.

Introduction

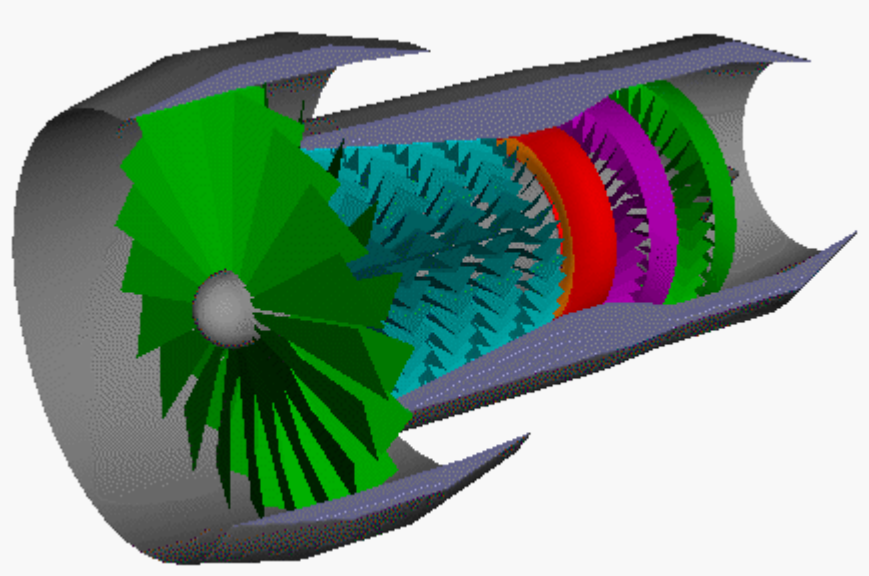


Problem Statement

Industries rely heavily on rotating machinery and engine systems.

Unexpected failures lead to:

- Production downtime
- High maintenance cost
- Safety risks



Operational inefficiency

There is a need to develop a system that:

- Detects abnormal operational behavior early
- Provides actionable alerts before failure
- Enables data-driven maintenance decisions
- Predicts Remaining Useful Life (RUL) of equipment

Objectives

Functional Objectives

1. Predict Remaining Useful Life (RUL) using machine learning.
2. Detect abnormal behavior using anomaly detection.
3. Provide early warning signals before system failure.
4. Visualize degradation trends through an interactive dashboard.

Technical Objectives

1. Implement Random Forest for RUL prediction.
2. Implement Isolation Forest for anomaly detection.
3. Perform feature engineering (rolling statistics on sensor data).
4. Build a deployable monitoring dashboard using Streamlit.

Non-Functional Objectives

1. Ensure scalable and modular design.
2. Maintain real-time responsiveness.
3. Provide clear and interpretable visual insights.
4. Deliver a user-friendly monitoring interface.

Scope

- **Applicable to** industrial rotating machinery and engine systems.
- **Designed for** predictive maintenance in manufacturing, aviation, and heavy industry.
- **Supports** data-driven maintenance planning instead of reactive repair.
- Can be extended to **real-time IoT sensor streams**.
- **Scalable** for multiple engines and large sensor datasets.
- Adaptable to different types of equipment with retraining.
- Can be integrated with enterprise monitoring systems.

Project Limitations

- Currently validated on simulated **NASA turbofan dataset**.
- Performance depends on quality and consistency of sensor data.
- Requires retraining for different machine types.

Dataset Description

Dataset Used

NASA CMAPSS Turbofan Engine Degradation Simulation Dataset

- Source: NASA Prognostics Data Repository
- Dataset: FD001 subset
- Type: Multivariate time-series sensor data
- Total Engines: 100 (training)
- Sensors per engine: 21 sensors
- Operational settings: 3

Why This Dataset?

- Industry-standard benchmark for predictive maintenance research.
- Simulates realistic degradation patterns.
- Widely used in academic and industrial studies.

Dataset Description

Dataset Used

Each record contains:

1. Engine ID
2. Cycle number
3. Operational settings
4. 21 sensor readings

Failure occurs when engine reaches its maximum cycle.

Remaining Useful Life (RUL) is computed as:

$$RUL = MaxCycle - CurrentCycle$$

Dataset Description

```
col_names = ['id','cycle'] + \
            [f'op_set_{i}' for i in range(1,4)] + \
            [f'sensor_{i}' for i in range(1,22)]
df = df.dropna(axis=1)
df.columns = col_names
```

```
df.head(15)
```

	id	cycle	op_set_1	op_set_2	op_set_3	sensor_1	sensor_2	sensor_3	sensor_4	sensor_5	...	sensor_12	sensor_13	sensor_14	sensor_15	sensor_16	sensor_17	sensor_18	sensor_19	sensor_20	sensor_21
0	1	1	-0.0007	-0.0004	100.0	518.67	641.82	1589.70	1400.60	14.62	...	521.66	2388.02	8138.62	8.4195	0.03	392	2388	100.0	39.06	23.4190
1	1	2	0.0019	-0.0003	100.0	518.67	642.15	1591.82	1403.14	14.62	...	522.28	2388.07	8131.49	8.4318	0.03	392	2388	100.0	39.00	23.4236
2	1	3	-0.0043	0.0003	100.0	518.67	642.35	1587.99	1404.20	14.62	...	522.42	2388.03	8133.23	8.4178	0.03	390	2388	100.0	38.95	23.3442
3	1	4	0.0007	0.0000	100.0	518.67	642.35	1582.79	1401.87	14.62	...	522.86	2388.08	8133.83	8.3682	0.03	392	2388	100.0	38.88	23.3739
4	1	5	-0.0019	-0.0002	100.0	518.67	642.37	1582.85	1406.22	14.62	...	522.19	2388.04	8133.80	8.4294	0.03	393	2388	100.0	38.90	23.4044
5	1	6	-0.0043	-0.0001	100.0	518.67	642.10	1584.47	1398.37	14.62	...	521.68	2388.03	8132.85	8.4108	0.03	391	2388	100.0	38.98	23.3669
6	1	7	0.0010	0.0001	100.0	518.67	642.48	1592.32	1397.77	14.62	...	522.32	2388.03	8132.32	8.3974	0.03	392	2388	100.0	39.10	23.3774
7	1	8	-0.0034	0.0003	100.0	518.67	642.56	1582.96	1400.97	14.62	...	522.47	2388.03	8131.07	8.4076	0.03	391	2388	100.0	38.97	23.3106
8	1	9	0.0008	0.0001	100.0	518.67	642.12	1590.98	1394.80	14.62	...	521.79	2388.05	8125.69	8.3728	0.03	392	2388	100.0	39.05	23.4066
9	1	10	-0.0033	0.0001	100.0	518.67	641.71	1591.24	1400.46	14.62	...	521.79	2388.06	8129.38	8.4286	0.03	393	2388	100.0	38.95	23.4694
10	1	11	0.0018	-0.0003	100.0	518.67	642.28	1581.75	1400.64	14.62	...	521.40	2388.01	8140.58	8.4340	0.03	392	2388	100.0	38.94	23.4787
11	1	12	0.0016	0.0002	100.0	518.67	642.06	1583.41	1400.15	14.62	...	521.80	2388.02	8134.25	8.3938	0.03	391	2388	100.0	39.06	23.3660
12	1	13	-0.0019	0.0004	100.0	518.67	643.07	1582.19	1400.83	14.62	...	521.85	2388.08	8128.10	8.4152	0.03	393	2388	100.0	38.93	23.2757
13	1	14	0.0009	-0.0000	100.0	518.67	642.35	1592.95	1399.16	14.62	...	521.67	2388.00	8134.43	8.3964	0.03	393	2388	100.0	39.18	23.3826
14	1	15	-0.0018	-0.0003	100.0	518.67	642.43	1583.82	1402.13	14.62	...	522.50	2388.08	8127.56	8.4199	0.03	391	2388	100.0	38.99	23.3500

15 rows × 26 columns

Technological Details

Programming Environment

- Python
- Pandas, NumPy, Matplotlib, Plotly
- Scikit-learn, Streamlit, Joblib

Machine Learning Models

- Random Forest Regressor – RUL prediction, handles nonlinear degradation and noise
- Isolation Forest – Anomaly detection, trained on healthy cycles to flag deviations

Deployment

- Streamlit dashboard with multi-engine visualization and real-time anomaly threshold tuning

Methodology

Step 1: Data Preprocessing

- Assigned column names, removed low-variance sensors
- Calculated and capped RUL for regression stability

Step 2: Feature Engineering

- Correlation-based sensor selection
- Rolling mean & std dev (window = 5), grouped engine-wise to prevent leakage

Step 3: Model Training

- Engine-level train-test split
- Linear Regression (baseline) → Random Forest (primary)
- Evaluated using RMSE & MAE

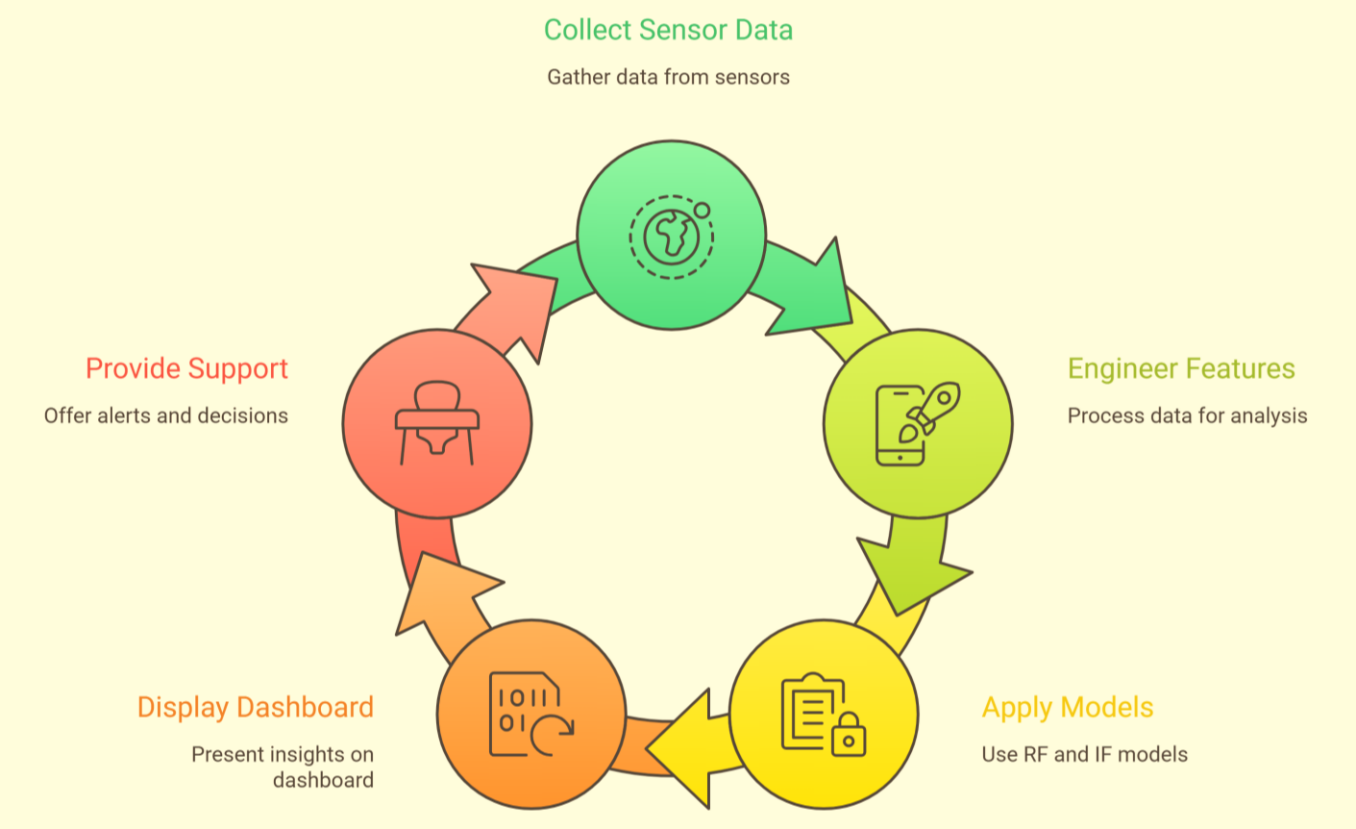
Step 4: Anomaly Detection

- Defined healthy region (RUL > 100)
- Trained Isolation Forest → generated anomaly scores & early warnings

Step 5: Deployment

- Serialized models via Joblib
- Streamlit dashboard visualizing RUL, anomaly scores, lead time & risk indicator

System Architecture



Dual-model framework combining supervised RUL prediction and unsupervised anomaly detection for robust failure forecasting.

Feature Engineering Strategy

Objective: To extract degradation patterns from raw sensor signals.

- Removed low-variance sensors
- Selected high-correlation degrading sensors
- Applied Rolling Mean (window = 5)
- Applied Rolling Standard Deviation
- Grouped operations engine-wise (prevent leakage)
- Applied RUL Capping for regression stability

Model Performance and Evaluation

Key Metrices

Model	RMSE	MAE
Linear Regression	21.17	17.03
Random Forest	17.90	12.85

- Random Forest significantly outperformed Linear Regression
- Nonlinear degradation captured effectively

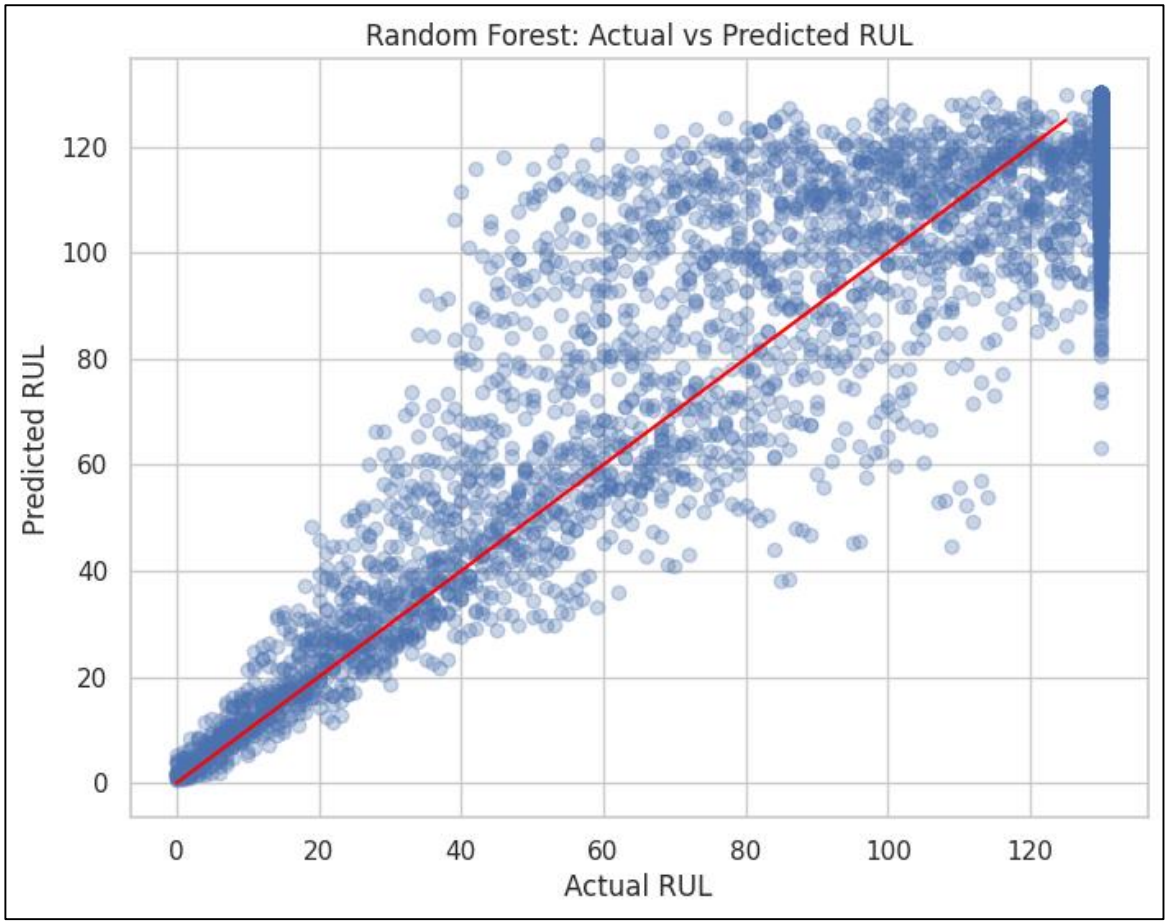
Random Forest reduced prediction error by ~15%, indicating nonlinear degradation behavior in engine sensors.

Model Performance and Evaluation

Scatterplot:
Actual RUL vs Predicted RUL
(Random Forest)

What it shows:
•Points clustered around diagonal
•Improved alignment compared to baseline
•Reduced spread at lower RUL

Label clearly:
•X-axis: Actual RUL
•Y-axis: Predicted RUL



Anomaly Detection & Early Warning Analysis

Healthy Training Logic

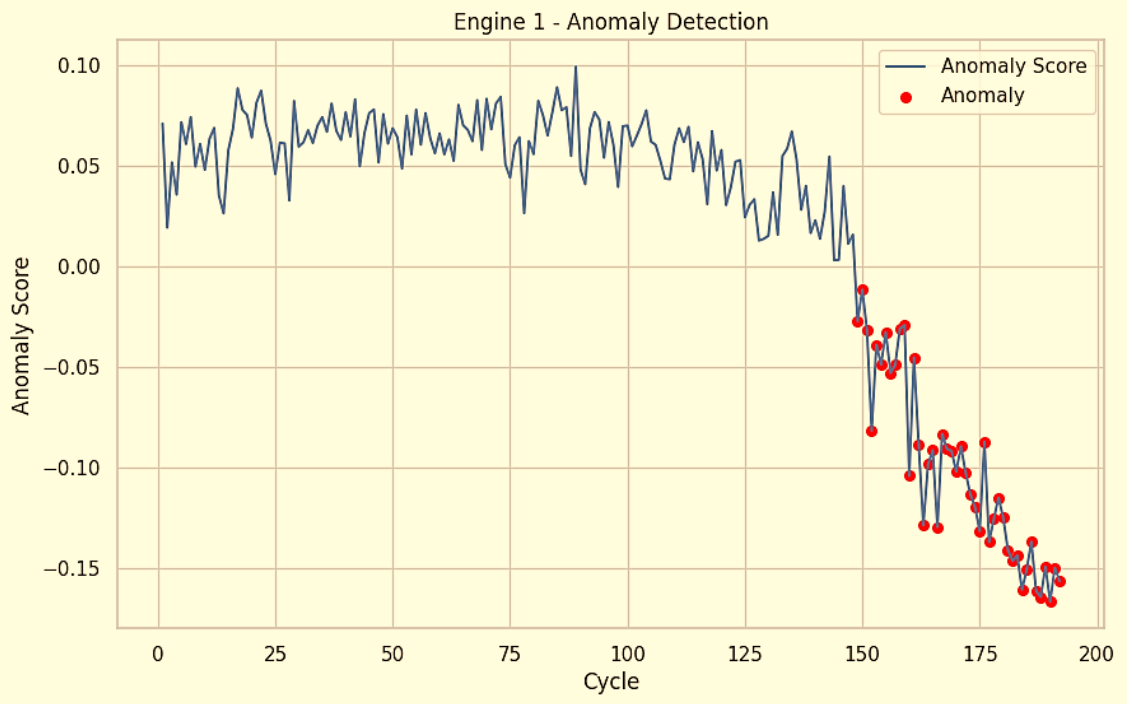
- Isolation Forest trained only on healthy cycles
- Healthy defined as: **RUL > 100**
- Model learns normal operational distribution

Anomaly Monitoring

- Generated anomaly score per cycle
- Lower score → higher abnormality
- Threshold-based risk classification

Early Warning Insight

First anomaly detected ~40 cycles before failure
Enables proactive maintenance planning
Provides redundancy alongside RUL prediction



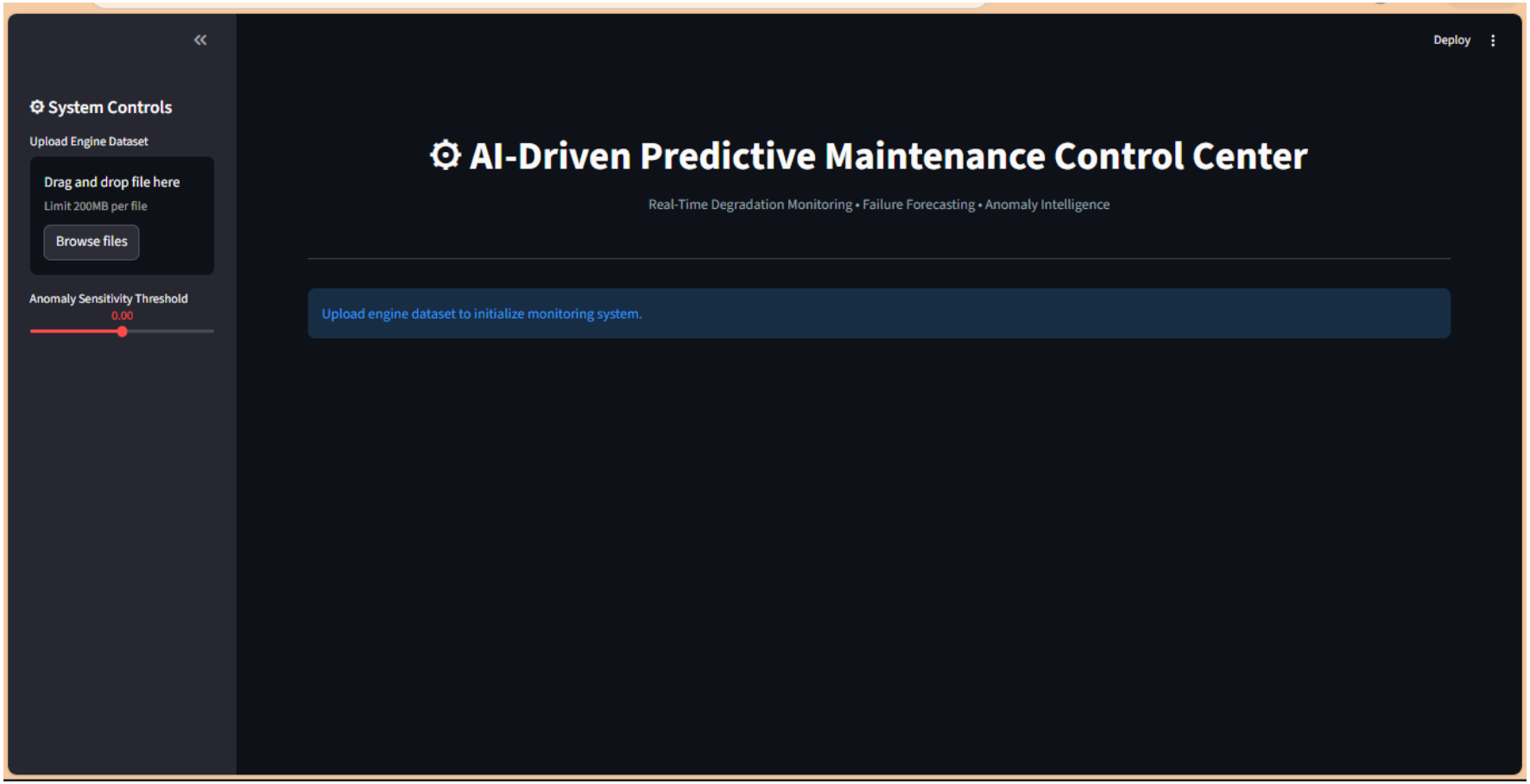
Weekly Work Details

Week	Technical Progress
Week 1	Dataset understanding, industrial maintenance problem framing, RUL formulation.
Week 2	Feature cleaning, variance filtering, time-series visualization.
Week 3	Rolling feature engineering, correlation analysis, leakage prevention strategy.
Week 4	Model training (LR & RF), hyperparameter tuning, performance benchmarking.
Week 5	Anomaly detection modeling, threshold tuning, lead-time analysis.
Week 6	End-to-end deployment pipeline, model serialization, dashboard development and testing.

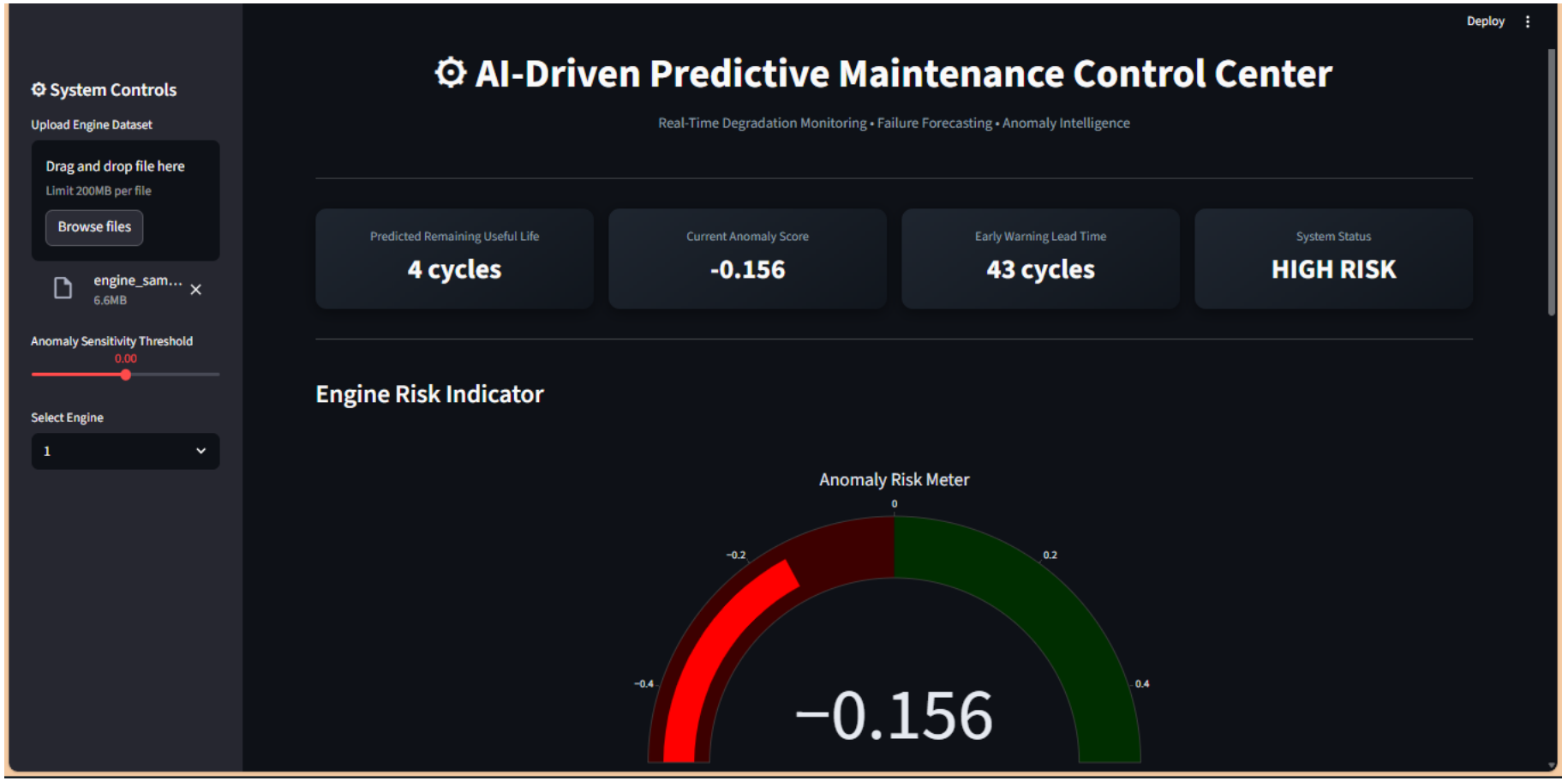
Milestones Achieved:

- Achieved RF RMSE ≈ 17.9
- Successfully detected anomalies ~40 cycles before failure
- Deployed interactive monitoring system

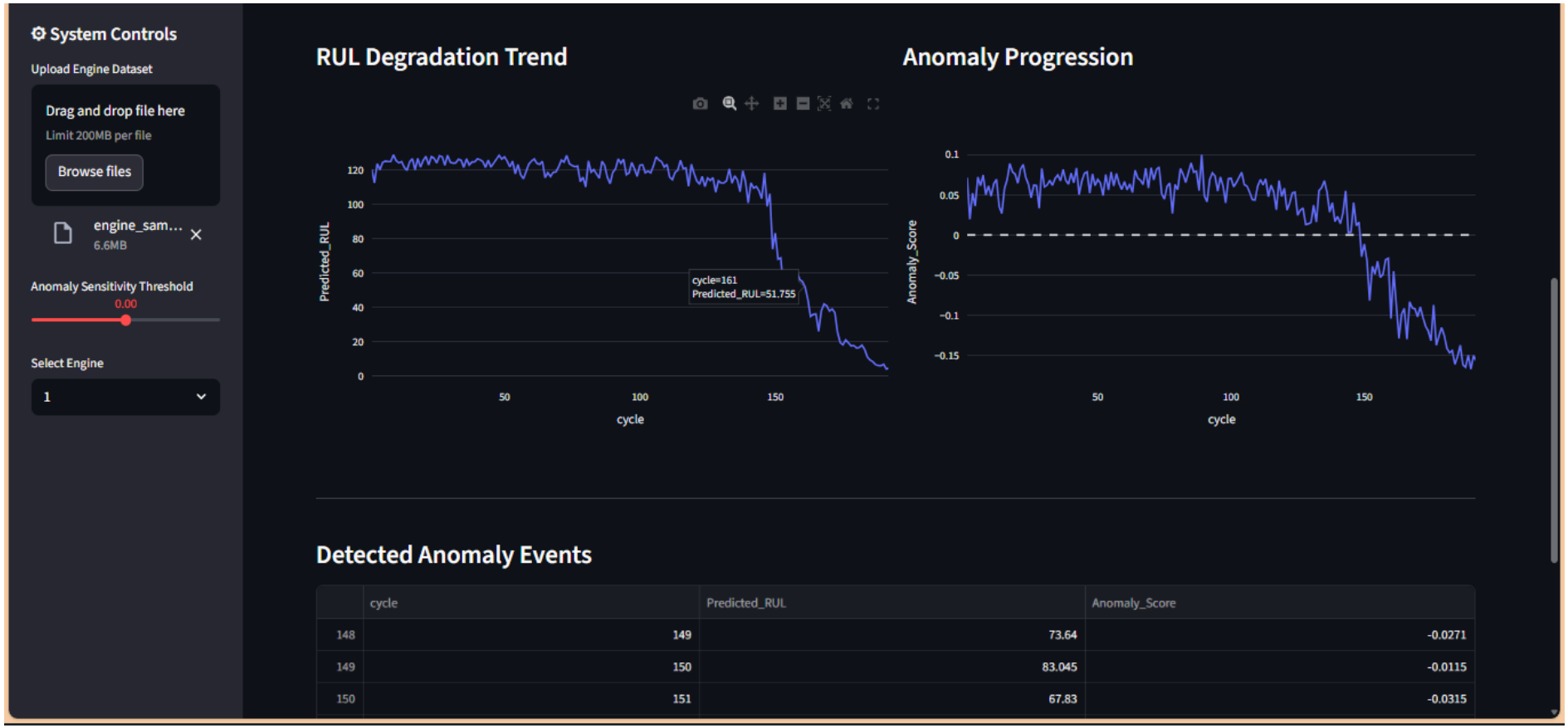
Results


System Controls
Upload Engine Dataset
Drag and drop file here
Limit 200MB per file
Browse files
Anomaly Sensitivity Threshold
0.00
Deploy
AI-Driven Predictive Maintenance Control Center
Real-Time Degradation Monitoring • Failure Forecasting • Anomaly Intelligence
Upload engine dataset to initialize monitoring system.

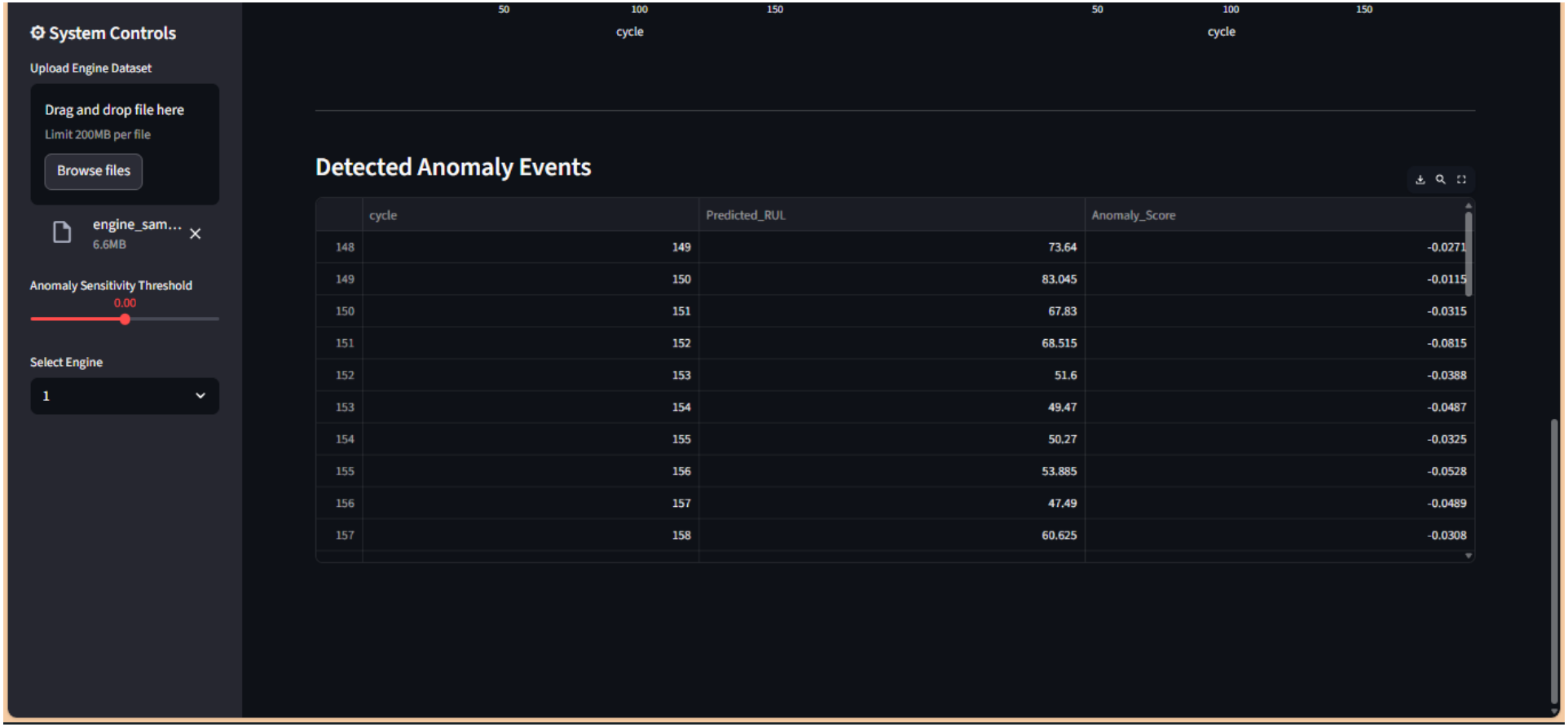
Results


Deploy
⚙ AI-Driven Predictive Maintenance Control Center
Real-Time Degradation Monitoring • Failure Forecasting • Anomaly Intelligence
⚙ System Controls
Upload Engine Dataset
Drag and drop file here
Limit 200MB per file
Browse files
engine_sam...
6.6MB
Anomaly Sensitivity Threshold
0.00
Select Engine
1
Predicted Remaining Useful Life
4 cycles
Current Anomaly Score
-0.156
Early Warning Lead Time
43 cycles
System Status
HIGH RISK
Engine Risk Indicator
Anomaly Risk Meter
0
-0.2
0.2
-0.4
0.4
−0.156

Results


System Controls
Upload Engine Dataset
Drag and drop file here
Limit 200MB per file
Browse files
engine_sam...
6.6MB
Anomaly Sensitivity Threshold
0.00
Select Engine
1
RUL Degradation Trend
Predicted_RUL
cycle
cycle=161
Predicted_RUL=51.755
Anomaly Progression
Anomaly_Score
cycle
Detected Anomaly Events
cycle
Predicted_RUL
Anomaly_Score
148 149 73.64 -0.0271
149 150 83.045 -0.0115
150 151 67.83 -0.0315

Results



	cycle	Predicted_RUL	Anomaly_Score
148	149	73.64	-0.0271
149	150	83.045	-0.0115
150	151	67.83	-0.0315
151	152	68.515	-0.0815
152	153	51.6	-0.0388
153	154	49.47	-0.0487
154	155	50.27	-0.0325
155	156	53.885	-0.0528
156	157	47.49	-0.0489
157	158	60.625	-0.0308

Conclusion

This project successfully developed an end-to-end AI-based predictive maintenance system, achieving strong RUL prediction performance with a Random Forest RMSE of approximately 17.9 and detecting early failure patterns nearly 40 cycles before breakdown. By combining supervised and unsupervised learning models, the system delivers robust and comprehensive equipment monitoring, further demonstrated through a fully interactive real-time dashboard deployment.

This system enables data-driven maintenance decisions instead of reactive repair strategies.

Future Scope

- Integration with real-time IoT sensor streams.
- Implementation of deep learning models (LSTM / GRU) for sequence modeling.
- Integration with ERP / CMMS systems for automated maintenance scheduling.
- Cloud-based deployment for scalable industrial monitoring.
- Adaptive threshold learning for dynamic anomaly detection.

The framework can evolve into a full-scale industrial asset health management platform.

Research and References

1. Saxena & K. Goebel, “NASA Turbofan Engine Degradation Simulation Data Set,” NASA Prognostics Data Repository.
2. S. M. Elsherif, “Deep Learning Prognostic Strategy for Remaining Useful Life”, Scientific Reports, 2025.
3. A. Motrani et al., Predictive Maintenance: A Machine Learning Approach to RUL, ICIMPE 2024.
4. K. Ensarioğlu et al., Deep Learning–Based RUL Estimation of Turbofan Engines, MDPI Applied Sciences, 2023.
5. “Isolation Forest,” Wikipedia.M. L. Ndao, Machine Learning Preprocessing & Explainability Impact, 2025.
6. B. Taşcı et al., RUL Prediction in Manufacturing Using ML Models, 2025.
7. NASA CMAPSS Dataset Description.Isolation Forest Survey – IEEE (various authors), 2025.


Asante
Gracias
धन्यवाद
Thankyou
Köszi
Suksema
Dankie
Ենդրսի
Ευχαριστώ
Danke
Ahéhee
Thank
Dîk
Tak
Dank
Jai-rruh-jef
ڰركشـ
Terima
kasih
Grazie
Merci
Спасибо
Ngiyabonga
Misaotra
Obrigado
ขอบคุณ